

$^{36}\text{Ar}(\text{p},\text{t})$ 2018Lo11,1972Pa02,1969Br21

$^{36}\text{Ar} \rightarrow 2\text{n} + ^{34}\text{Ar}$ from $J^\pi=0^+$ ^{36}Ar ground state.

2018Lo11: a 100-MeV proton beam was produced from the RCNP WS beam line at Osaka University. The target was 0.76-mg/cm² enriched ^{36}Ar gas ($\geq 99\%$). Reaction products were momentum-analyzed by the Grand Raiden spectrograph and detected using the focal-plane detector system. Measured $\sigma(E_t, \theta)$. Deduced levels. Calculated astrophysical $^{30}\text{S}(\alpha, \text{p})^{33}\text{Cl}$ reaction rate. Also see **2009ObZY**.

1972Pa02: 40-45-MeV proton beams were produced from the MSU sector-focused cyclotron. The target was $>99\%$ -enriched ^{36}Ar gas. Reaction products were detected using a detector telescope made up of two silicon surface-barrier transmission mounted counters. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, J, and π from JULIE-DWBA analysis.

1969Br21: a 45-MeV proton beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas. Reaction products were detected using a 155- μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

1974Ha02: a 40.2-MeV proton beam was produced from the MSU cyclotron. Measured Q value. Tritons for the g.s. to g.s. transition measured using split-pole magnetic spectrograph, FWHM=10-25 keV. Measures Q value= -19513 3. Deduced mass excess of ^{34}Ar .

 ^{34}Ar Levels

E(level) [†]	J^π [‡]	L [‡]	Comments
0	0 ⁺	0	
2091.6 24	2 ⁺	2	E(level): weighted average of 2094 11 (1972Pa02) and 2091.5 24 (2018Lo11). Other: 2097 20 (1969Br21). L: 2 in 1972Pa02 .
3290.9 18	2 ⁺	2	E(level): weighted average of 3303 25 (1969Br21), 3288 14 (1972Pa02), and 3290.9 18 (2018Lo11). L: 2 in 1972Pa02 .
3879.1 33	0 ⁺	0	E(level): weighted average of 3899 25 (1969Br21), 3879 15 (1972Pa02), and 3878.8 33 (2018Lo11). L: 0 in 1972Pa02 for unresolved 3879+4050 and 3879 is more strongly excited.
4035 16			E(level): unweighted average of 4050 14 (1972Pa02) and 4019.1 43 (2018Lo11).
4515.1 23	(3 ⁻)	(3)	E(level): weighted average of 4522 14 (1972Pa02) and 4514.9 23 (2018Lo11). Other: 4560 40 (1969Br21). L: not fitted particularly well by either L=2 or L=3 calculated distributions. However, it is distinctly different from the other experimental L=2 distributions. There is also a 3 ⁻ state at approximately the same excitation energy in the mirror nucleus ^{34}S (1969Br21).
4641.5 21			E(level): weighted average of 4651 14 (1972Pa02) and 4641.3 21 (2018Lo11).
4875.3 38			E(level): weighted average of 4867 14 (1972Pa02) and 4875.9 38 (2018Lo11).
4967.9 27	0 ⁺	0	E(level): weighted average of 4970 40 (1969Br21), 4985 14 (1972Pa02), and 4967.2 27 (2018Lo11).
5262 15			
5317 13			E(level): weighted average of 5340 40 (1969Br21), 5307 13 (1972Pa02), and 5330 17 (2018Lo11).
5535 18			
5629.6 45			
5929 20			E(level): unweighted average of 5909 12 (1972Pa02) and 5948.2 66 (2018Lo11).
6050.1 36	(2 ⁺)	(2)	E(level): weighted average of 6074 11 (1972Pa02), and 6049.2 20 (2018Lo11). Other: 6100 40 (1969Br21). L: (2) assigned by evaluators. The total angular distribution of unresolved 5909+6074 exhibits some L=2 character, and therefore possibly one or both states are 2 ⁺ , but no definite assignment can be made (1972Pa02). L=2 is assigned for 6100 in 1969Br21 .
6525 9			
6641 14			
6723 22			
6807 4			E(level): weighted average of 6794 11 (1972Pa02) and 6808 3 (2018Lo11). Other: 6860 40 (1969Br21).
6917 5			
7072 8			
7276 2			
7340 11	2 ⁺	2	E(level): unweighted average of 7340 50 (1969Br21), 7322 6 (1972Pa02), and 7358 4 (2018Lo11).
7502 16	(2 ⁺)	(2)	E(level): unweighted average of 7530 50 (1969Br21), 7499 4 (1972Pa02), and 7476 2 (2018Lo11).

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 $^{36}\text{Ar}(\text{p,t})$ [2018Lo11](#), [1972Pa02](#), [1969Br21](#) (continued)

 ^{34}Ar Levels (continued)

$E(\text{level})^\dagger$	Comments
7921 18	E(level): unweighted average of 7950 50 (1969Br21), 7925 5 (1972Pa02), and 7889 2 (2018Lo11).
8166 3	
8660 9	
8746 8	
8899 8	
9004 4	
9148 22	
9226 18	
9549 16	

[†] From [2018Lo11](#), unless otherwise noted.

[‡] From [1969Br21](#), unless otherwise noted. Selection rules: for the (p,t) reaction on 0^+ targets, the total transferred angular momentum $J = L_{2n}$ and the parity $\pi = (-1)^{L_{2n}}$.